

# Inference for Multiple and Logistic Regression

IMS Chapters 25, 9 (background), 26

## Part 1: Multiple regression inference (Ch. 25)

## From one predictor to many

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

Everything from last time carries over:

- ▶ each  $b_i$  has a standard error
- ▶  $T = \frac{b_i - 0}{SE_i}$ , now with  $df = n - k - 1$
- ▶ CI:  $b_i \pm t_{df}^* \times SE_i$
- ▶ same LINE conditions

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**What changes is the meaning of every hypothesis.**

Each test is *conditional*

For predictor  $x_i$ :

$H_0 : \beta_i = 0$  given all the other variables in the model

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So a small p-value says:  $x_i$  *adds information beyond what the other predictors already provide.*

A large p-value does **not** say “ $x_i$  is unrelated to  $y$ .” It may be redundant with something else in the model.

## Example: loan interest rates

```
loans_full_schema: interest_rate ~ debt_to_income + term +  
credit_checks
```

| term           | estimate | std.error | statistic | p.value  |
|----------------|----------|-----------|-----------|----------|
| (Intercept)    | 4.31     | 0.20      |           | < 0.0001 |
| debt_to_income | 0.04     | 0.00      | 13.3      | < 0.0001 |
| term           | 0.16     | 0.00      | 37.9      | < 0.0001 |
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*Interpret the `credit_checks` row, including the hypothesis being tested.*

Holding debt-to-income and term fixed, each additional credit check goes with an average **0.25 percentage point** increase in interest rate. The test rejects  $\beta = 0$  given the other two variables are already in the model.

## Interpretation requires “holding others fixed”

The slope on `credit_checks` is **not** the slope you would get from a simple regression of `interest_rate` on `credit_checks`. It is the slope *after accounting for* the other predictors.

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Same variable, different models, **different coefficients and different p-values**.  
Which brings us to...

## Multicollinearity

## Collinear predictors

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**A coin-dish example** (invented data,  $n = 26$ ): predict the dish's total value from the number of coins, and the number of *low-value* coins.

| model                     | coefficient(s)   | p-value             |
|---------------------------|------------------|---------------------|
| number_of_coins alone     | 0.13             | $< 0.0001$          |
| number_of_low_coins alone | 0.02             | 0.7                 |
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- ▶ number\_of\_low\_coins looked useless alone, but is highly significant *with* number\_of\_coins.
- ▶ Its coefficient even **flips sign**.

## What multicollinearity does

- ▶ Coefficients and SEs become unstable; small data changes can change signs.
- ▶ Individual p-values can be misleading (each variable looks redundant given the other).
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### Practical rules

- ▶ Look at the correlation between predictors before interpreting coefficients.
- ▶ Interpret *sets* of related variables together, or drop one.
- ▶ Never read a single coefficient as “the effect of  $x_i$ ” when predictors are tangled.

Choosing a model

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**Prediction goal:** forecast new observations. Use **cross-validation**: hold out data, predict it, and compare errors. P-values are not the main tool.

## Cross-validation example: penguins

Predict `body_mass_g` for 4 folds, comparing:

- ▶ **Small model:** `bill_length_mm` only ( $\hat{y} = 362.3 + 87.4x$ )
- ▶ **Large model:** adds `bill_depth_mm`, `flipper_length_mm`, `sex`, `species`

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The small model's CV error is about **5.1 times** the large model's. The extra variables really help prediction.

## Data snooping

If you try many models or many predictors and keep the best-looking, your p-values are too optimistic.

- ▶ Decide on the question and model **before** looking at the outcome when you can.
- ▶ Treat a surprising finding in a huge search as a **hypothesis**, not a result.
- ▶ Cross-validation on **held-out** data protects against overfitting.

## Part 2: Logistic regression background (Ch. 9)

## Why a new model?

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If we let  $y \in \{0, 1\}$  and fit a line, predictions can land below 0 or above 1. We need a model for a **probability**.

## The logit link

Let  $p$  = probability that  $y = 1$ . Model the **log-odds**:

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k = \eta$$

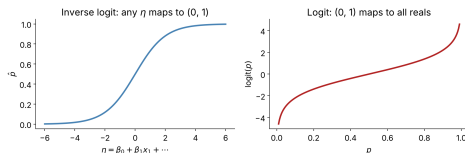
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Solve for  $p$  (the inverse logit):

$$p = \frac{e^\eta}{1 + e^\eta}$$



Any value of  $\eta$  maps to a valid probability.

## Example: resumes and callbacks

Bertrand and Mullainathan sent resumes to employers, randomly assigning names signalling race and sex. Response: `received_callback`.

Model with `honors` only:

$$\log \left( \frac{\hat{p}}{1 - \hat{p}} \right) = -2.4998 + 0.8668 \times \text{honors}$$

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*Find the predicted callback probability for a resume **without** honors, and **with** honors.*

- ▶ No honors:  $\eta = -2.4998 \Rightarrow \hat{p} = \frac{e^{-2.4998}}{1 + e^{-2.4998}} \approx 0.076$
- ▶ Honors:  $\eta = -2.4998 + 0.8668 = -1.633 \Rightarrow \hat{p} \approx 0.163$

## Interpreting a coefficient

- ▶ **Sign:** positive means higher probability.
- ▶ **Size:** a one-unit increase in  $x_i$  changes the **log-odds** by  $\beta_i$ , which multiplies the **odds** by  $e^{\beta_i}$ .
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For honors:  $e^{0.8668} \approx 2.4$ . Holding other variables fixed, having honors multiplies the odds of a callback by about 2.4.

## Many predictors and AIC

The full model for `resume` had **AIC = 2677**.

**Backward elimination:** remove the variable whose removal lowers AIC the most; stop when no removal helps.

- ▶ Dropping `college_degree` gives  $AIC = 2676$ , which is better, so drop it.
- ▶ Lower AIC is better, with a penalty for each added parameter.

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Reduced model:

| term              | estimate |
|-------------------|----------|
| (Intercept)       | -2.72    |
| job_city: Chicago | -0.44    |
| years_experience  | 0.02     |
| honors            | 0.76     |
| military          | -0.34    |
| has_email_address | 0.22     |
| race: White       | 0.44     |
| sex: Male         | 0.20     |

## A prediction

White man, Chicago, 14 years' experience, no honors, no military, has email:

$$\hat{\eta} = -2.72 - 0.44 + 0.02(14) + 0.22 + 0.44 - 0.20 \approx -2.4$$

$$\hat{p} = \frac{e^{-2.4}}{1 + e^{-2.4}} \approx 0.08 \quad \Rightarrow \quad \text{about 12 applications per callback}$$

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The same profile for a Black man has  $\hat{p} \approx 0.054$ , about 18 applications per callback.

*This is a prediction from the model. Is the race effect statistically significant?*

*That is an **inference** question, which is next.*

## Part 3: Inference for logistic regression (Ch. 26)

# Conditions

For logistic regression we need:

1. **Independence:** each outcome is independent of the others (given the predictors).
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The model-based tests and intervals also lean on large samples (Wald z-tests).

## Wald tests for each coefficient

Same recipe, now with a **z** statistic (not  $t$ ):

$$Z = \frac{b_i - 0}{SE_i}, \quad H_0 : \beta_i = 0 \text{ given the other variables}$$

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$$Z = \frac{b_i - 0}{SE_i}, \quad H_0 : \beta_i = 0 \text{ given the other variables}$$

Confidence interval for a coefficient:  $b_i \pm 1.96 \times SE_i$ .

For an **odds ratio**, exponentiate the endpoints:

$$(e^{b_i - 1.96 SE_i}, e^{b_i + 1.96 SE_i})$$

## Example: spam email

email data, 3921 emails. Response: spam.

| term        | estimate | std.error | z    | p-value  |
|-------------|----------|-----------|------|----------|
| (Intercept) | -2.05    | 0.06      |      |          |
| to_multiple | -1.91    | 0.30      | -6.4 | < 0.0001 |
| cc          | 0.02     | 0.02      | 1.0  | 0.245    |
| dollar      | -0.07    | 0.02      | -3.5 | 0.0005   |
| urgent_subj | 2.66     | 0.80      | 3.3  | 0.0009   |

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to\_multiple, dollar, and urgent\_subj are significant. cc is not (given the other three).

## Intervals for odds ratios

urgent\_subj:  $b = 2.66$ ,  $SE = 0.80$

$$2.66 \pm 1.96(0.80) = (1.09, 4.23)$$

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An urgent subject line multiplies the odds of spam by about 14 (plausibly anywhere from 3 to 69), holding the other predictors fixed.

to\_multiple:  $e^{-1.91} \approx 0.15$ . Emails sent to multiple people have about **one seventh** the odds of being spam.

## Checking the model: diagnostics

Does  $\hat{p} = 0.2$  really correspond to 20% spam?

1. Bucket the predicted probabilities (e.g., 0–0.1, 0.1–0.2, ...).
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Systematic gaps suggest the model is missing something (non-linearity, interactions, a variable).

## P-values or prediction?

Same two goals as before.

- ▶ **Inference:** Wald tests, intervals, odds ratios.
- ▶ **Prediction:** use **cross-validation accuracy** (e.g., true-positive rate for spam, true-positive rate for non-spam).

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At a 10% cutoff, a one-variable model (`to_multiple`) catches nearly all spam (about 96–98%) but has overall accuracy only about 23–26%, because it flags too much. A 7-variable model reaches accuracy about 76–80% with spam caught about 65–75% of the time.

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**A model can be significant and still be a poor predictor**, and a good predictor can contain non-significant variables.

Wrap-up

## Summary: three regression models

|            | Simple linear   | Multiple linear            | Logistic                         |
|------------|-----------------|----------------------------|----------------------------------|
| Response   | numeric         | numeric                    | yes / no                         |
| Test stat  | $t, df = n - 2$ | $t, df = n - k - 1$        | $z$                              |
| $H_0$      | $\beta_1 = 0$   | $\beta_i = 0$ given others | $\beta_i = 0$ given others       |
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- ▶ Inference is **conditional** on the other variables in the model.
- ▶ **Multicollinearity** destabilizes individual coefficients.
- ▶ **Prediction** and **inference** are different goals with different tools.
- ▶ Simulation (`infer`) and math models give the same story when conditions hold.